Lab 2: Relational Algebra Practice

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This document contains the solutions for the Practice exercises (Section 4) in the Lab 2 guide, using the RelaX calculator based on the Course Management ERD.
- Find the average salary of employees in each department.
- List the names of employees who don't work on any projects.
- List the names of all employees and their dependents.
- List the names of employees who are also managers of departments.
- Find the names of all employees who work on every project.
- Find the names of employees with the same supervisor as the employee with social security number '123456789'.

Exercise 1: Find the average salary of employees in each department.

Relational Algebra Expression:

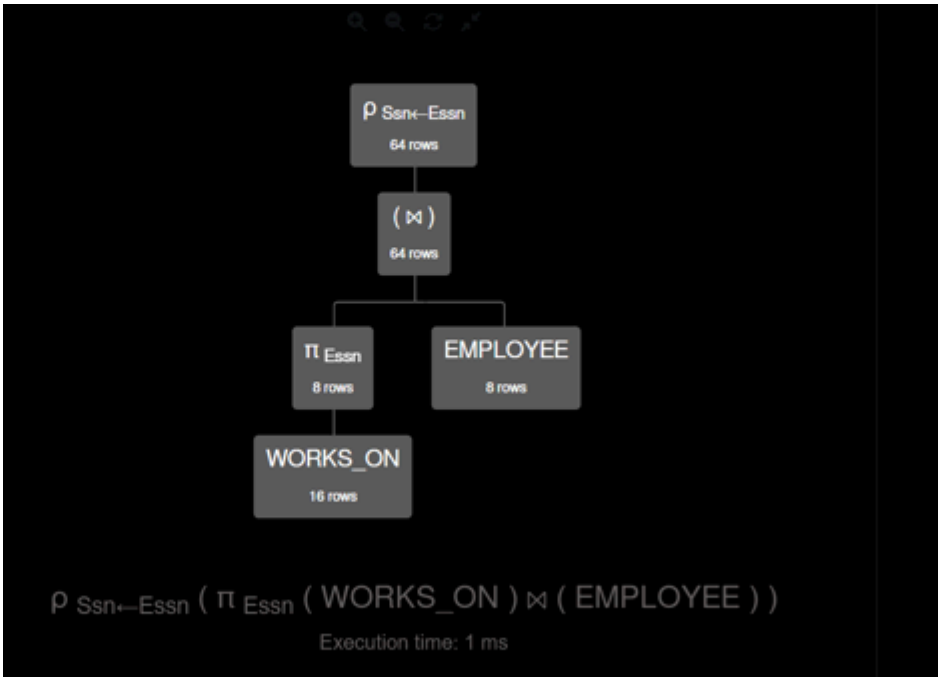
```
ɣ Dno ← avg(Salary) (EMPLOYEE)
```

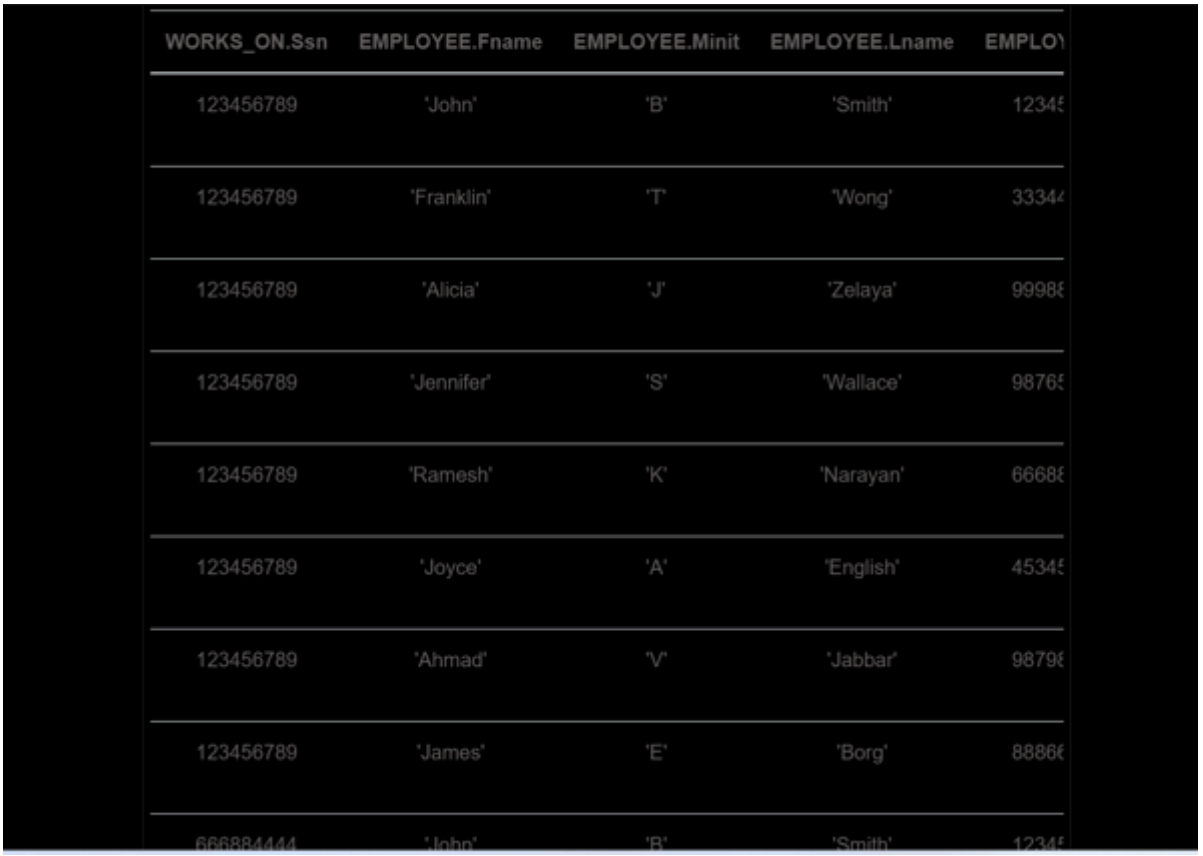


Exercise 2: List the names of employees who don't work on any projects.

Relational Algebra Expression:

`ρ Ssn<-Essn (π Essn (WORKS_ON) ⋈ (EMPLOYEE))`



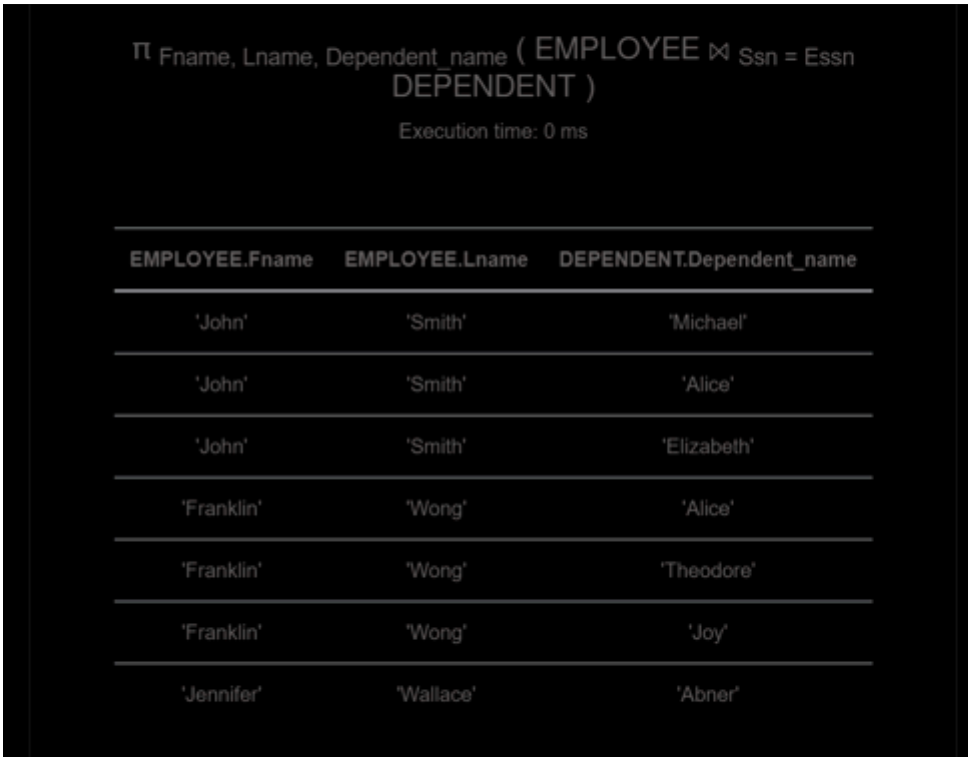


WORKS_ON.Ssn	EMPLOYEE.Fname	EMPLOYEE.Minit	EMPLOYEE.Lname	EMPLOY
123456789	'John'	'B'	'Smith'	1234
123456789	'Franklin'	'T'	'Wong'	3334
123456789	'Alicia'	'J'	'Zelaya'	9998
123456789	'Jennifer'	'S'	'Wallace'	9876
123456789	'Ramesh'	'K'	'Narayan'	6668
123456789	'Joyce'	'A'	'English'	4534
123456789	'Ahmad'	'V'	'Jabbar'	9879
123456789	'James'	'E'	'Borg'	8886
666884444	'John'	'B'	'Smith'	1234

Exercise 3: List the names of all employees and their dependents.

Relational Algebra Expression:

```
π Fname, Lname, Dependent_name (EMPLOYEE ⋈ Ssn=Essn DEPENDENT)
```



EMPLOYEE.Fname	EMPLOYEE.Lname	DEPENDENT.Dependent_name
'John'	'Smith'	'Michael'
'John'	'Smith'	'Alice'
'John'	'Smith'	'Elizabeth'
'Franklin'	'Wong'	'Alice'
'Franklin'	'Wong'	'Theodore'
'Franklin'	'Wong'	'Joy'
'Jennifer'	'Wallace'	'Abner'

Exercise 4: List the names of employees who are also managers of departments.

Relational Algebra Expression:

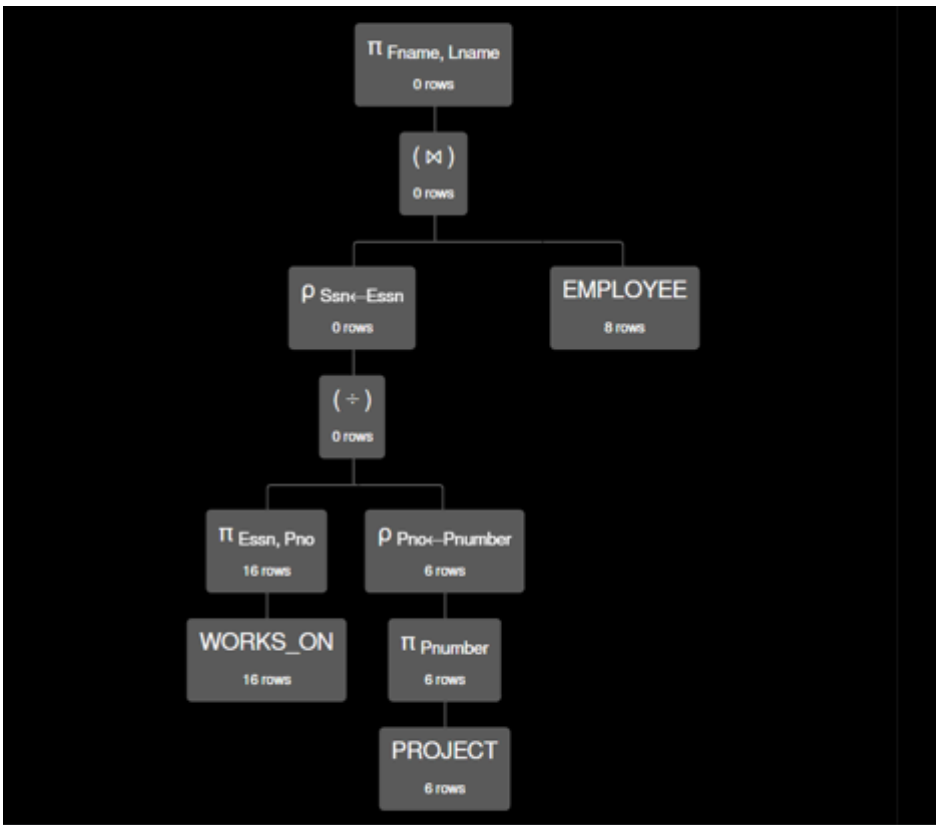
```
π Fname, Lname (EMPLOYEE ⋈ Ssn=Mgr_ssn DEPARTMENT)
```

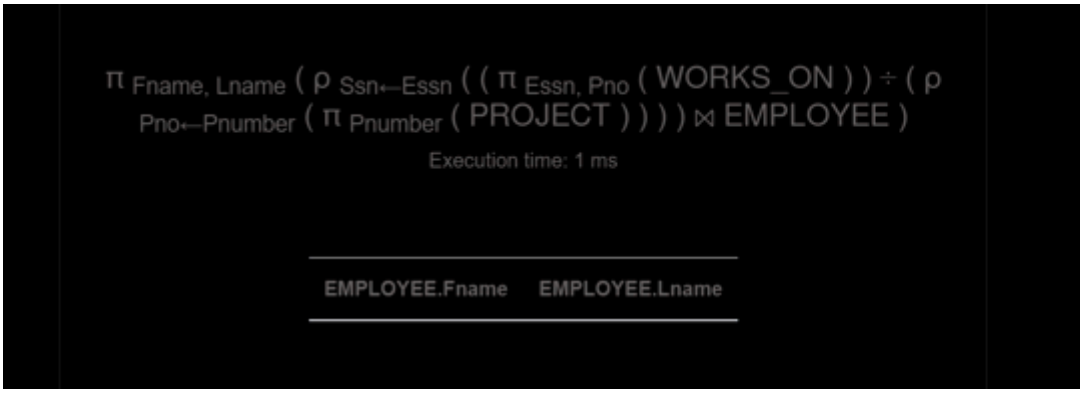


Exercise 5: Find the names of all employees who work on every project.

Relational Algebra Expression:

```
π Fname, Lname (ρ Ssn<-Essn ((π Essn, Pno (WORKS_ON)) ÷ (ρ Pno<-Pnumber (π Pnumber (PROJECT)))) ⋈ EMPLOYEE)
```

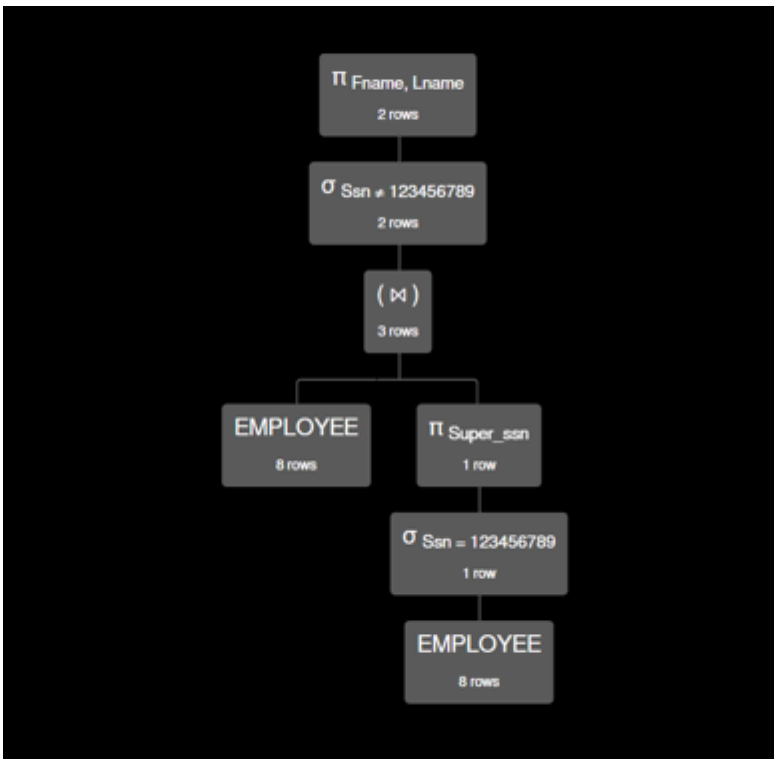




Exercise 6: Find the names of employees with the same supervisor as the employee with social security number '123456789'

Relational Algebra Expression:

```
π Fname, Lname (σ Ssn != 123456789 (EMPLOYEE ⋈ (π Super_ssn (σ Ssn = 123456789 (EMPLOYEE)))))
```



π Fname, Lname (σ Ssn ≠ 123456789 (EMPLOYEE ⋈ (π Super_ssn (σ Ssn = 123456789 (EMPLOYEE)))))

Execution time: 4 ms

EMPLOYEE.Fname	EMPLOYEE.Lname
'Ramesh'	'Narayan'
'Joyce'	'English'